

Econ 101A

Section 5 (Solutions)

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We can now find an optimum. This section is about *using* it. Two questions. First, how does the optimal choice move when a parameter (a price, a wage) changes, even when we cannot solve for it in closed form? The tool is the **Implicit Function Theorem**. Second, what is the Lagrange multiplier really? We will *derive* that it equals $\partial V/\partial b$ with the envelope theorem, and build an intuition for it as a shadow price.

1 Comparative Statics with the Implicit Function Theorem

Recall the IFT: if $F(x; p) \equiv 0$ defines x as a function of p , then, wherever $F_x \neq 0$,

$$\frac{dx}{dp} = -\frac{F_p}{F_x}.$$

The trick. The optimal choice $x^*(p)$ is *defined* by the first-order condition. For $\max_x f(x; p)$, the optimum solves $f_x(x^*(p); p) = 0$. This is exactly an equation $F(x; p) = 0$ with $F = f_x$. Apply the IFT:

$$\frac{dx^*}{dp} = -\frac{f_{xp}}{f_{xx}}.$$

The second-order condition says $f_{xx} < 0$ at a maximum, so the denominator is negative and

$$\text{sign}\left(\frac{dx^*}{dp}\right) = \text{sign}(f_{xp}).$$

In words: the optimal action rises with a parameter exactly when that parameter raises the *marginal* payoff of the action. This one line is the engine behind most comparative-statics results, and it never required solving the FOC.

Two choice variables. With $\max_{x,y} f(x, y; p)$, differentiate both FOCs in p to get a linear system whose coefficient matrix is the Hessian H_f ; solving it (Cramer's rule) gives dx^*/dp and dy^*/dp . Same idea, more bookkeeping.

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2 The Envelope Theorem, Derived

Let $V(b) = \max_{x,y} f(x,y)$ subject to $g(x,y) = b$, with solution $(x^*(b), y^*(b), \lambda(b))$. We show $V'(b) = \lambda$.

Since $V(b) = f(x^*(b), y^*(b))$, the chain rule gives

$$V'(b) = f_x x^{*'} + f_y y^{*'}.$$

At the optimum the FOCs say $f_x = \lambda g_x$ and $f_y = \lambda g_y$, so

$$V'(b) = \lambda(g_x x^{*'} + g_y y^{*'}).$$

Now differentiate the constraint $g(x^*(b), y^*(b)) = b$ with respect to b :

$$g_x x^{*'} + g_y y^{*'} = 1.$$

Substituting,

$$\boxed{V'(b) = \lambda}.$$

The indirect effects through x^* and y^* collapsed because at the optimum the objective and the constraint move together. Shortcut: $V'(b) = \partial \mathcal{L} / \partial b|_{\text{opt}} = \lambda$, since $\mathcal{L} = f + \lambda(b - g)$.

3 What a Multiplier Is

- **A shadow price.** λ is the extra objective you get per extra unit of the constrained resource: the internal price the optimizer would pay for one more unit. In the consumer's problem $b = m$, so $\lambda = \partial V / \partial m$ is the **marginal utility of income**.
- **A common marginal value.** At the optimum, the last unit of the resource yields the *same* marginal payoff, λ , wherever you spend it. If one use paid more, you would move a unit there; the fact that you stopped means the margins are equal. λ is that common margin.
- **Everywhere.** A budget gives the marginal utility of income; a firm's output target gives marginal cost; an hour budget gives the value of an hour; a capacity limit gives the rent you would pay for one more unit of capacity.

4 Exercises

Exercise 1 (IFT: signing a firm's input demand)

A competitive firm chooses an input $x \geq 0$ to maximize profit $\pi(x) = pF(x) - wx$, where $F'(x) > 0$ and $F''(x) < 0$, and $p, w > 0$.

- Write the first-order condition, and explain why for a general F you cannot solve it in closed form.
- Use the IFT to sign $\frac{dx^*}{dw}$ and $\frac{dx^*}{dp}$.
- Interpret, and say where the second-order condition entered.

Solution. (a) FOC: $\pi'(x) = pF'(x) - w = 0$, i.e. the marginal revenue product equals the wage. For a general concave F you cannot invert F' to get a formula for x^* .

(b) Write the FOC as $\Phi(x^*; w, p) = pF'(x^*) - w = 0$. Differentiating,

$$\frac{dx^*}{dw} = -\frac{\Phi_w}{\Phi_x} = -\frac{-1}{pF''(x^*)} = \frac{1}{pF''(x^*)} < 0, \quad \frac{dx^*}{dp} = -\frac{\Phi_p}{\Phi_x} = -\frac{F'(x^*)}{pF''(x^*)} > 0,$$

using $F'' < 0$ (so the denominator is negative) and $F' > 0$.

(c) Input use falls when the wage rises and rises when the output price rises. The SOC $pF'' < 0$ is exactly what fixes the signs; without concavity the comparative statics would be ambiguous.

Exercise 2 (derive $\lambda = \partial V / \partial m$)

A consumer maximizes $u(x, y) = \alpha \ln x + \beta \ln y$ subject to $p_x x + p_y y = m$ (with $\alpha, \beta > 0$).

- Recall the demands and the multiplier from Section 3.
- Write the indirect utility $V(p_x, p_y, m) = u(x^*, y^*)$ and show it has the form $(\alpha + \beta) \ln m + C(p_x, p_y)$, where C does not depend on m .
- Compute $\partial V / \partial m$ and check it equals λ .
- Interpret λ and give its units.

Solution. (a) From Section 3: $x^* = \frac{\alpha}{\alpha + \beta} \frac{m}{p_x}$, $y^* = \frac{\beta}{\alpha + \beta} \frac{m}{p_y}$, and $\lambda = \frac{\alpha + \beta}{m}$.

$$(b) V = \alpha \ln \left(\frac{\alpha}{\alpha + \beta} \frac{m}{p_x} \right) + \beta \ln \left(\frac{\beta}{\alpha + \beta} \frac{m}{p_y} \right) = (\alpha + \beta) \ln m + \underbrace{\alpha \ln \frac{\alpha}{(\alpha + \beta)p_x} + \beta \ln \frac{\beta}{(\alpha + \beta)p_y}}_{C(p_x, p_y), \text{ no } m}.$$

$$(c) \frac{\partial V}{\partial m} = \frac{\alpha + \beta}{m} = \lambda. \checkmark$$

(d) λ is the marginal utility of income: one extra dollar raises maximized utility by about λ . Units: utils per dollar. (Notice it falls as m rises: income is worth less at the margin when you already have more.)

Exercise 3 (the multiplier as a common marginal value)

You have a fixed budget B to split between two activities with concave returns $R_1(x_1)$ and $R_2(x_2)$ (each $R'_i > 0$, $R''_i < 0$): $\max_{x_1, x_2} R_1(x_1) + R_2(x_2)$ subject to $x_1 + x_2 = B$.

- Set up the Lagrangian and show the optimum equalizes marginal returns: $R'_1(x_1^*) = R'_2(x_2^*) = \lambda$.
- What does λ say about the value of an extra unit of budget?
- Suppose at some allocation $R'_1(x_1) > R'_2(x_2)$. Which way should you move a unit of budget, and why?

Solution. (a) $\mathcal{L} = R_1(x_1) + R_2(x_2) + \lambda(B - x_1 - x_2)$. FOCs: $R'_1(x_1) - \lambda = 0$ and $R'_2(x_2) - \lambda = 0$, so $R'_1(x_1^*) = R'_2(x_2^*) = \lambda$: at the optimum the last unit spent on each activity earns the same marginal return.

(b) By the envelope theorem $\frac{dV}{dB} = \lambda$: one more unit of budget raises the maximized total return by about λ , the common marginal return.

(c) If $R'_1(x_1) > R'_2(x_2)$, a unit of budget earns more in activity 1, so move it from 2 to 1. Because returns are concave, doing so lowers R'_1 and raises R'_2 ; keep going until they meet at λ .